Lecture #19: Limit Behavior and Doubly Stochastic Chains

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Motivation

Example (Ehrenfest chain)

- We have two urns with balls in them.
- At the start of the next round, we randomly select a ball between the two urns and move it to the other urn.
- Let X_n be the number of balls in the "left" urn after the n^{th} draw.

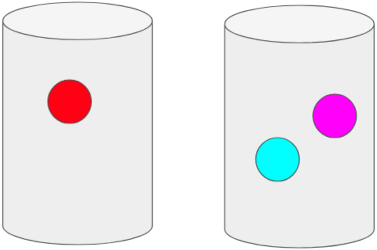
When $N = 3$, we get

In the second power of p the zero pattern is shifted:

Question

Will this persist?

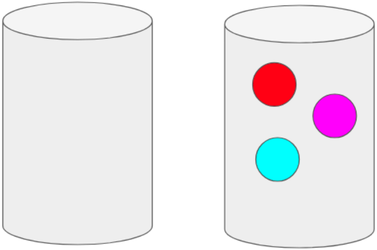
Initial Urn Set-up: Odd number of balls in the left urn



	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 1: Let's consider this the set-up to the problem, i.e. $X_0 = 1$ since there is one ball in the left urn.

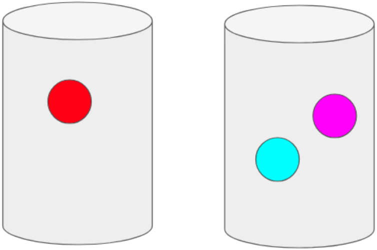
One possible next step: Even number of balls in the left urn



	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 2: $\frac{1}{3}$ of the time, the only ball in the left urn will be selected and moved to the right urn.

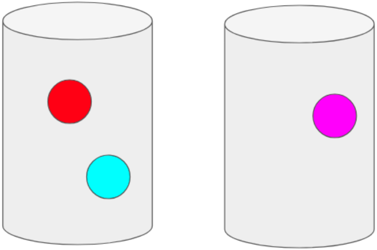
Back to the Initial Urn Set-up: Odd number of balls in the left urn



	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 3: With probability 1, we return back to the initial set up after two steps.

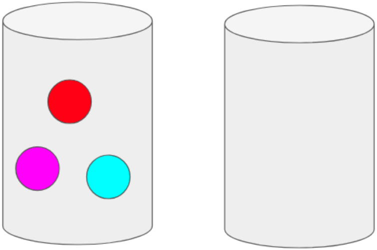
The other possible next step: Even number of balls in the left urn



	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 4: $\frac{2}{3}$ of the time, one ball in the right urn will be selected and moved to the left urn.

The next step: Odd number of balls in the left urn



	0	1	2	3
0	0	3/3	0	0
1	1/3	0	2/3	0
2	0	2/3	0	1/3
3	0	0	3/3	0

Figure 5: $\frac{1}{3}$ of the time, one ball in the right urn will be selected and moved to the left urn.

Ehrenfest chain Summary

- This alternation between even and odd means that it is impossible to be back where we started after an odd number of steps. You need an even number of steps to get back.
- In the second power of p the zero pattern is shifted:

$$p^2 = \begin{array}{c|cccc} & \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{0} & 1/3 & 0 & 2/3 & 0 \\ \mathbf{1} & 0 & 7/9 & 0 & 2/9 \\ \mathbf{2} & 2/9 & 0 & 7/9 & 0 \\ \mathbf{3} & 0 & 2/3 & 0 & 1/3 \end{array}$$

- In symbols, if n is odd then

Periodicity

Definition

The period of a state is the largest number that will divide all the $n \geq 1$ for which $p^n(x,x) > 0$. That is, it is the greatest common divisor of $I_x = \{n \geq 1 : p^n(x,x) > 0\}$.

Remark

To check that this definition works correctly, we note that in the Ehrenfest exmaple,

Playing with Periodicity

Example (Triangle and Square)

	−2	**−1**	**0**	**1**	**2**	**3**	
−2	0	0	1	0	0	0	0
−1	1	0	0	0	0	0	0
0	0	0.5	0	0.5	0	0	0
1	0	0	0	0	0	1	0
2	0	0	0	0	0	0	1
3	0	0	1	0	0	0	0

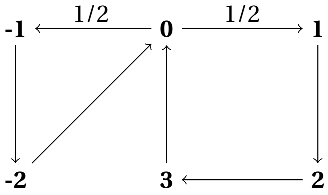
- from zero, equally likely to gor to 1 or -1.
- If we go -1:
  - we go to -2, then
  - we go back to 0.

Playing with Periodicity

Example (Triangle and Square)

	−2	**−1**	**0**	**1**	**2**	**3**	
−2	0	0	1	0	0	0	0
−1	1	0	0	0	0	0	0
0	0	0.5	0	0.5	0	0	0
1	0	0	0	0	0	1	0
2	0	0	0	0	0	0	1
3	0	0	1	0	0	0	0

- from zero, equally likely to gor to 1 or -1.
- If we go -1:
  - we go to -2, then
  - we go back to 0.
- If we go 1:
  - we go to 2, then
  - we go to 3, then
  - we go back to 0.
- The name refers to the fact that $0 \to -1 \to -2 \to 0$ is a triangle and $0 \to 1 \to 2 \to 3 \to 0$ is a square.



Question

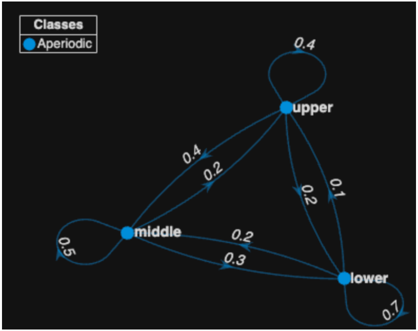
What are the possible periods of this problem? What is I_0?

The Reality of Periodicity

While periodicity is a theoretical possibility, it rarely manifests itself in applications, except occasionally as an odd-even parity problem, e.g., the Ehrenfest chain.

Definition (Aperiodic)

In most cases we will find (or design) our chain to be *aperiodic*, i.e., all states have period 1.



Theory of aperiodicity

Lemma

If $p(x, x) > 0$, then x has period 1.

Proof.

$\square$

Lemma

If $\rho_{xy} > 0$ and $\rho_{yx} > 0$ then x and y have the same period.

Lemma

If $\rho_{xy} > 0$ and $\rho_{yx} > 0$ then x and y have the same period.

Remark

The short answer is that if the two states have different periods, then by going from x to y, from y to y in the various possible ways, and then from y to x, we will get a contradiction.

The Holy Grail

Theorem (Convergence Theorem)

Let p be irreducible, aperiodic, and there is a stationary distribution π. Then as $n \to \infty$, $p^n(x, y) \to \pi(y)$.

Example: Social Mobility

Example

	Lower	**Middle**	**Upper**
Lower	0.7	0.2	0.1
Middle	0.3	0.5	0.2
Upper	0.2	0.4	0.4

Question

Is it irreducible?

Question

Is it aperiodic?

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Doubly Stochastic Transition Matrices

Definition (Doubly Stochastic)

A transition matrix p is said to be *doubly stochastic* if its COLUMNS sum to 1, or in symbols

Remark

Remember this is different from our Markov matrices ("singlely stochastic") which only have their rows sum to 1,

The Stationary Distribution for doubly stochastic matrices is uniform!

Theorem

If p is a doubly stochastic transition probability for a Markov chain with N states, then the uniform distribution, $\pi(x) = 1/N$ for all x, is a stationary distribution.

Proof.

To figure this out, we need to remember that $\pi p = \pi$ is the formula for a stationary distribution. If we assume $\pi(x) = 1/N$, then

□

Examples of Doubly Stochastic Matrices

Example (Symmetric random walk (with bumpers))

Let's say the states are $\{0, 1, 2, ..., L\}$. The chain goes to the right or left at each step with probability $1/2$. However, if you are at 0 or L, and try to go out of bounds, the bumpers keep you at the same state.

	0	**1**	**2**	**2**	**2**
0	0.5	0.5	0	0	0
1	0.5	0	0.5	0	0
2	0	0.5	0	0.5	0
3	0	0	0.5	0	0.5
4	0	0	0	0.5	0.5

Question

What is the stationary distribution?

Tiny Board Game

Consider a circular board game with only six spaces $\{0, 1, 2, 3, 4, 5\}$. On each turn we roll a die with 1 on three sides, 2 on two sides, and 3 on one side to decide how far to move.

Probability Transition Matrix of "Tiny Board Game"

Question

Is this doubly stochastic?